



Who benefits most from structured management practices? Evidence from the World Bank enterprise survey

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Abstract

A significant body of research assumes that modern management practices are less relevant for smaller firms, especially in developing countries. We analyze the relationship between structured management practices and firm performance using World Bank Enterprise Survey (WBES) data covering 94 countries. The positive relationship between structured management practices and firm sales, often documented for larger firms in developed countries, is also present for smaller firms in developing economies. This association is stronger for smaller firms. This points to a resolution to a persistent question in the economics of management literature: if modern management techniques are so good, why don't more firms use them? Our findings suggest the answer is not low marginal benefits; the marginal benefits are highest among smaller firms. Instead, barriers to adoption such as implementation and regulatory costs may be more relevant, consistent with prior work by Bloom (American Economic Review 109:1648–1683, 2019).

Keywords World Bank enterprise survey · Developing countries · Firm performance · Small and medium enterprises

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1 Introduction

A large body of evidence documents that “structured management practices” are positively associated with performance among larger firms in developed countries (Bloom and Van Reenen, 2007; Bloom et al., 2012a; Broszeit et al., 2019; Ohlsbom and Maliranta, 2021; Englmaier et al., 2023; Lamorgese et al., 2024). Research on developing countries or small firms remains scarce (Scur et al., 2021). This oversight has been dismissed as unimportant. Bloom et al. (2016), for example, assert that “the formal management practices we consider will not be so important for smaller firms” (p.12). Likewise, McKenzie and Woodruff (2017) believe that “Management in the small-firm context is less focused on human relations than is the case in larger firms” (p. 2967). This paper tests these assertions and finds them to be lacking.

Small firms contribute significantly to employment and output in most countries (Di Bella et al., 2023). However, lessons learned in one context may not be profitably transplanted to different contexts. This may result in wasted opportunities or even economic losses (Quinn & Scur, 2021).

This paper fills a gap in the literature by examining how structured management practices correlate with firm performance across a diverse sample of firms in countries of differing levels of economic, institutional, and social development. To examine for this, we use the Enterprise Surveys conducted by the World Bank (henceforth WBES), which began gathering data on management practices and firm characteristics in 2018.

We find that firms with higher management scores have higher sales, even after adjusting for firm characteristics and controlling for country-wide characteristics, firm economic activity, and year-fixed effects. Management practices and sales are positively correlated and statistically significant in low and lower-middle-income countries. Furthermore, even small firms can benefit significantly from employing structured management practices.

Management practices vary across countries and cultures. For instance, Bloom et al. (2012a) report that firms in the United States are more likely to use financial incentives, whereas firms in Japan, Sweden, and Germany prefer monitoring and target management. Given this variation, a natural question is whether the effects of these practices also vary across contexts.

This paper contributes to the literature in three ways. First, few recent studies examine the economic effects of structured management practices across a large sample of countries and firms. The closest research is Bloom et al. (2012 a, b), which focus mainly on firms in developed countries. We extend this work with more recent data spanning 94 countries at varying levels of development. Second, we find an economically significant positive association between overall management score and firm performance for firms of all sizes, but especially among smaller sized firms. Third, the association between management practices and sales is as strong in low-income countries as in high-income countries.

The remainder of the paper is organized as follows: Sects. 2 and 3 present the theoretical framework and review the related literature. Section 4 describes the main variables of interest and the estimation methodology. Section 5 provides the empirical results, showing the relationship between structured management practices for

firms of different sizes and in different economic contexts. Section 6 discusses some limitations of the study. Section 7 concludes by summarizing the main findings of the paper.

2 Conceptual framework: the resource-based view

This study draws on the Resource-Based View of the firm (Barney, 1991), which emphasizes how firm-specific resources shape strategy and performance. Within this framework, management practices function as key resources: valuable, rare, and hard to imitate. Good management practices resist imitation even with external consultants (Gerhart & Feng, 2021), and their implementation requires substantial investment (Jewell et al., 2022). When adopted, structured management practices can enhance operational efficiency and provide sustained competitive advantage. How valuable are these practices to firms, given the costs involved in their implementation? Does this value vary across firm types and economic contexts?

The academic literature offers several typologies of management practices. Bloom et al. (2012a) identified four broad categories: monitoring, incentivizing performance, setting targets, and managing operations. Bloom et al. (2019) simplified this to incentive versus non-incentive practices. Both types, when well designed and deployed appropriately, can help firms manage resources and improve profitability.

Drawing on the Resource-Based View and the existing empirical literature, we test the following hypotheses:

Hypothesis 1 *Structured management practices are positively associated with firm performance, as measured by sales revenue.*

Hypothesis 2 *Structured management practices are positively associated with firm performance among firms in developing countries.*

Hypothesis 3 *Structured management practices are positively associated with firm performance among small firms.*

Hypothesis 4 *The positive association between structured management practices and firm performance is stronger for smaller firms than for larger firms.*

3 Literature review

Persistent differences in firm performance and growth rates across and within countries have led economists to search for explanations beyond the usual hard technologies: patents, equipment, and machinery (Bloom & Van Reenen, 2010; Van Reenen, 2013; Scur et al., 2021). Van Reenen (2013) suggests that adopting management practices is a proximate cause of these differences. Giorcelli (2021) identifies three ways in which management affects performance. First, it facilitates identifying and adopting new production techniques that can help firms reach the technological

frontier. Second, management acts as a technology that improves allocating existing resources and capabilities. Third, effective management enhances a firm's human capital. Management is then an intangible asset that elevates a firm's output.

The lack of data on firms' adoption of effective management practices and a methodology to measure these practices had hindered research on the relationship between management practices and firm performance. The introduction of the World Management Survey (WMS) in the early 2000s, along with a reliable methodology for measuring structured management practices, has spurred research into the relationship between good management and various measures of firm performance.

Several papers use the WMS, or surveys with questionnaires like the WMS (Bloom and Van Reenen, 2007; Bloom et al., 2012b, 2022; Broszeit et al. 2019; Grover and Karplus, 2020; Cornwell et al., 2021; Kawasaki et al., 2023; Cao and Yong, 2024; Lamorgese et al., 2024). This strand of research primarily investigates the association between good management and various outcome measures among organizations.

Bloom and Van Reenen (2007) was the first study to construct a measure of good management practices, identify their determinants, and explore their association to firm performance. The sample included medium-sized manufacturing firms from the United States, the United Kingdom, France, and Germany. Using a standard augmented production function, the authors found a strong positive association between the management index and firm performance. Similarly, Bloom et al. (2012b) showed a connection between higher management scores and higher operating revenue using a sample of firms from Central Asian and Central European transition countries, as well as Germany and India. Likewise, Bloom et al. (2022) studied approximately eighteen thousand Mexican firms, finding that both value added per worker and firm performance increased with a higher management index. This body of evidence motivates Hypothesis 1: that structured management practices are positively associated with firm performance. While these studies provide robust evidence for developed economies, they leave open the question of whether these relationships extend to developing countries, motivating Hypothesis 2.

Since the management index typically consists of four dimensions—operations, monitoring, targets, and incentives—researchers also investigated the links between each dimension and overall firm performance (Bloom et al., 2012b; Broszeit et al., 2019; Grover and Karplus, 2020; Cao and Yong, 2024), finding mostly positive but varied results.

These research findings are suggestive but can make no causal claims as endogeneity is not accounted for. Reverse causality may occur, where firms with high productivity tend to adopt better management practices, for example, by hiring superior management consultants (Bloom & Van Reenen, 2007).

Bianchi and Giorcelli (2022) used a historical episode, the Training Within Industry (TWI) management program carried out in the United States during the 1940's to assess the impact of exogenously assigned management training programs on over eleven thousand firms' performance. The measured effects were substantial and long-lasting.

This literature focuses almost exclusively on medium and large firms in developed economies. The implicit assumption that small firms do not benefit from structured management practices remains untested. This gap motivates Hypotheses 3 and 4:

that the positive association extends to small firms, and that this association may be stronger for small firms than larger ones. Our paper provides a complementary and updated analysis of smaller firms in developing and developed countries. The literature shows that management practices complement other firm assets like human capital and training. We empirically examine the individual effect of those variables and add other factors that theoretically are important in explaining sales (Hatch & Dyer, 2004; Antonioli & Della Torre, 2016; Bergh et al., 2024).

4 Data and descriptive statistics

The data come from the World Bank Enterprise Surveys (WBES), conducted from 2018 to 2023. This survey gathers comprehensive insights into firms' operations, including ownership and governance structures, performance, export activities, age, number of employees, and R&D activity. Conducted through nationally representative, stratified samples across manufacturing, retail, and service sectors, the WBES provides a robust dataset for empirical analysis (The World Bank, 2023).

The large number of peer-reviewed articles using the WBES supports its relevance and reliability. Since 1999, over 600 peer-reviewed articles have used the WBES to study corruption, financial inclusion and constraints, gender gaps, firm performance and innovation, and other relevant subjects. To our knowledge, only one working paper has exploited this rich dataset to directly examine the effect of management practices on firm performance for a large number of developed and developing countries (Grover & Karplus, 2020).

4.1 Data on structured management practices

There is a large variation in management practices across firms and countries (Bloom & Van Reenen, 2010; Bloom et al., 2016, 2019; Scur et al., 2021). Those studies used the WMS dataset, which covers medium and large firms in developed countries, with data typically ending in 2016 (Scur et al., 2021).

The WBES, on the other hand, includes data for countries at very dissimilar stages of development. This study thereby extends the earlier literature by incorporating a richer cross-section of countries. Further, we divide the countries into high-income, upper-middle-income, and lower-middle and low-income countries based on the World Bank's classifications (see Appendix Table A3 for the list of countries in each category).

We use nine questions from the WBES to calculate scores on four critical dimensions of structured management practices: operations, monitoring, targets, and incentives. Breaking down the SMP score into these four components allows for a detailed analysis of the different facets of management practices within the surveyed firms.

We construct our SMP scores using an approach like that in Bloom et al. (2019) and other subsequent studies (Broszeit et al., 2019; Bloom et al., 2022; Grover & Karplus, 2020). On each of the management questions, the WBES restricts answers to a finite set of ordinal choices. These choices are re-coded to be equally spaced on a range of 0 to 1. For example, one question asks: "Who was aware of the production

targets at this establishment?” Four ordered options are allowed: Only senior managers (given a score of 0); Most managers and some production workers (given a score of 0.33); Most managers and most production workers (0.67); and all managers and most production workers (given a score of 1). Table A1 presents the wording of the nine questions in the WBES that we used and describes the scoring scale we used to calculate each measure of SMP.

Management practices are frequently categorized into four dimensions: monitoring, incentives, targets, and operations (Bloom & Van Reenen, 2007; Broszeit et al., 2019; Grover & Karplus, 2020; Kawasaki et al., 2023; Cao & Yong, 2024). Splitting management practices into four dimensions allows us to investigate whether different types of management practices have different effects on firm performance across groups of country incomes and sizes.

Each of the four dimensions or components of SMP (management, incentives, targeting, and operations management) is scored as the average of the answers to questions relating to it. The final SMP score is then the average of these four components. When reporting regression analysis, we use a further normalization where the SMP and its four components are standardized to be $N(0,1)$.

4.2 Distribution of firm management practices

Figure 1 provides up-to-date information on the distributions of management scores for firms in each of the three groups of countries.¹ Fig. 1 illustrates the significant heterogeneity in management practices across the entire sample and within the three income groups and confirms previous research findings regarding moderate skewness (Bloom & Van Reenen, 2010; Bloom et al., 2016, 2019; Scur et al., 2021). Management scores have nearly identical distributions for firms in Upper-Middle Income and those in High Income countries. The distribution of management scores for firms in Low and Low-Middle are shifted noticeably leftward. Firms in developing countries tend to use structured management practices less than firms in richer countries. The analysis in Sect. 5 explores whether this is due to fewer benefits from using SMPs vs. higher costs of implementation.

Table 1 provides complementary evidence of the high variance in the adoption of good management overall as well as for its components. The table reports the SMP scores, and the component scores, for the whole sample as well as categorizations based on firm-size or the country's development status.

A quick look at the data reveals some intriguing facts. On average, for the whole sample, the structured management score was 0.571. There is a positive correlation between a country's income and their average firm SMP scores. Firms in high-income countries report more adherence to SMP best practices (with a score of 0.613) than upper-middle countries (0.598), which in turn are higher than scores in low and lower-middle income countries (0.532).

For the whole sample, the averages of the four SMP components differ greatly. For instance, firms report more adherence to operations management (with an average

¹ The management practice scores in this section are measured on a 0–1 scale. To aid in interpretation, the econometric sections below use the z-scores.

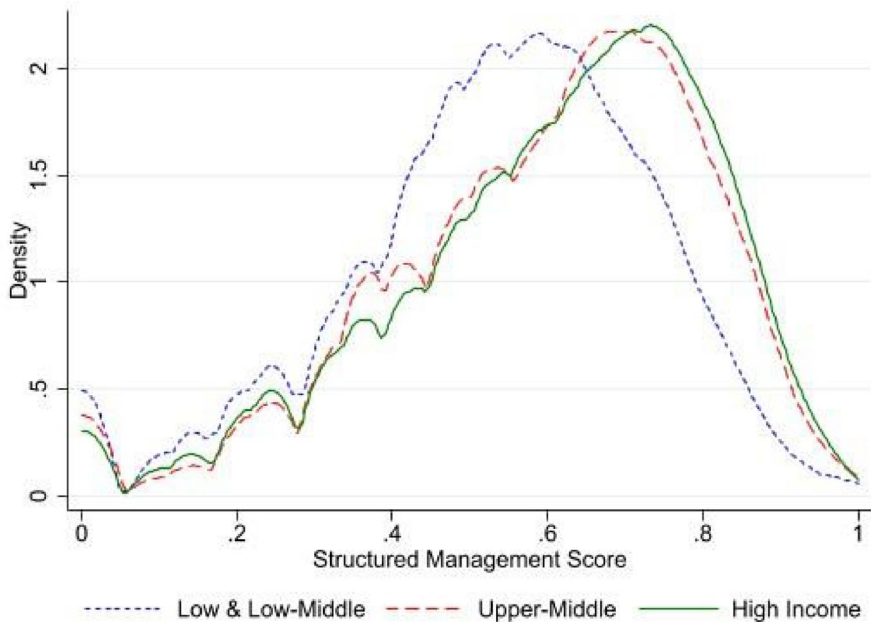


Fig. 1 Distribution of structured management scores for three national income groups

score of 0.727) and monitoring (0.629), than they do incentives (0.529) and targeting (0.490). The standard deviations across all individual components are quite large; there is a lot of heterogeneity in adopting each of the management practices.

One of the main focuses of this study is to analyze how the effect of SMP varies by firm size. The WBES classifies firms as small (5–19 employees), medium (20–99 employees), or large (100 or more employees) (The World Bank, 2023). According to this classification, small firms represent 46%, medium-sized firms account for 34%, and large firms make up the remaining 20% of the overall sample.

Figure 2 offers a view on the distributions of management scores across the three groups of firm sizes. The figure illustrates the large amount of variability in management scores within firms of all sizes. The distributions of raw management scores are also all left-skewed. Small firms show the least degree of skewness. More importantly, the distributions for small firms are shifted to the left whereas large firms are shifted to the right. Larger firms have less variance, and use SMP more than do smaller firms, which are more highly varied in their use of SMP, but tend to use it less.

Table 1 provides more numerical details on this relationship. As firms increase in size, management scores also increase. This relationship is consistent whether viewed as overall SMP, or across the four components of SMP. In fact, there is a positive and monotonic relationship between firm size and SMP scores; larger firms employ all facets of SMP more than do medium firms. Smaller firms employ SMP the least of all. This once again invites the question: Is this because the benefits to SMP are lower for smaller firms, or that the costs of implementation are higher? We

Table 1 Management scores across income groups, firm sizes, and sectors

Indicator	Income group				Firm size		
	All	Low & Low M	Upper Mid	High	Small	Med	Large
SMP (Total)	0.571 (0.195)	0.532 (0.191)	0.598 (0.191)	0.613 (0.191)	0.501 (0.207)	0.564 (0.191)	0.622 (0.177)
Monitoring	0.629 (0.235)	0.568 (0.228)	0.643 (0.239)	0.686 (0.223)	0.528 (0.213)	0.606 (0.223)	0.689 (0.238)
Incentives	0.529 (0.320)	0.533 (0.332)	0.556 (0.302)	0.508 (0.311)	0.448 (0.349)	0.521 (0.317)	0.583 (0.296)
Targeting	0.490 (0.315)	0.445 (0.292)	0.509 (0.325)	0.542 (0.330)	0.409 (0.329)	0.488 (0.319)	0.540 (0.289)
Operations	0.727 (0.269)	0.660 (0.295)	0.777 (0.222)	0.793 (0.231)	0.681 (0.280)	0.727 (0.264)	0.754 (0.265)

answer this question shortly, but first we must describe our dataset and the estimation methodology.

4.3 Summary statistics of other variables of interest and control variables

Firm performance, and whether to adopt effective management practices, can vary due to firm characteristics, industry factors, and national contexts (Bloom & Van Reenen, 2010; Bloom et al., 2016). Key factors influencing this variation include firm size, age, presence of foreign capital, export capability, and the skill of human capital. Accordingly, this study includes the following variables:

- (a) Sales: the value of sales revenue for each firm, in constant 2015 US dollars. For countries with a rapidly rising inflation rate, using a yearly exchange rate and inflation adjustment is problematic. Yearly values were converted from local currency using exchange rates derived from the World Bank's World Development Indicators. Observations with exchange rates below 0.0001 were set to missing as these are too noisy to be informative. Extreme values (below 1st or above 99th percentile) were set to missing. Finally, nominal values were converted to 2015 constant US dollars using the US's GDP deflator.
- (b) Employees: the total number of permanent full-time equivalent workers employed by the firm in the fiscal year (adjusted for temporary workers by the WBES).
- (c) Capital: the replacement value of all currently used capital, whether owned or rented, in 2015 US dollars.
- (d) Material: the total annual cost, in 2015 US dollars, of raw materials and intermediate goods used in production.
- (e) Age: the number of years the firm has been operating.
- (f) Manager experience: the number of years of experience working in the current sector that the top manager has.
- (g) Exporting: a dummy variable indicating whether the firm has any sales abroad.
- (h) Training: a dummy variable indicating whether the firm has formal training programs for its permanent, full-time employees.
- (i) Public: a dummy variable indicating whether a company has shares that are publicly traded in the stock market.
- (j) R&D: a dummy variable indicating whether the firm reported spending on research and development activities, either in-house or contracted with other companies, excluding market research surveys?
- (k) Foreign firm: a dummy variable indicating whether 10% or more of the firm is owned by foreign individuals, companies, or organizations.
- (l) Industry: a set of dummy variables corresponding to 2-digit industry codes from the (ISIC Rev. 4). This data is provided as part of the WBES and forms the basis of the industry fixed effects.

Table A2 gives further details on how each of these variables is constructed from the items in the WBES.

Table 2 reports summary statistics of the firm-level variables used in the empirical section of this study. The mean and median size of firms was 136 and 50 full-

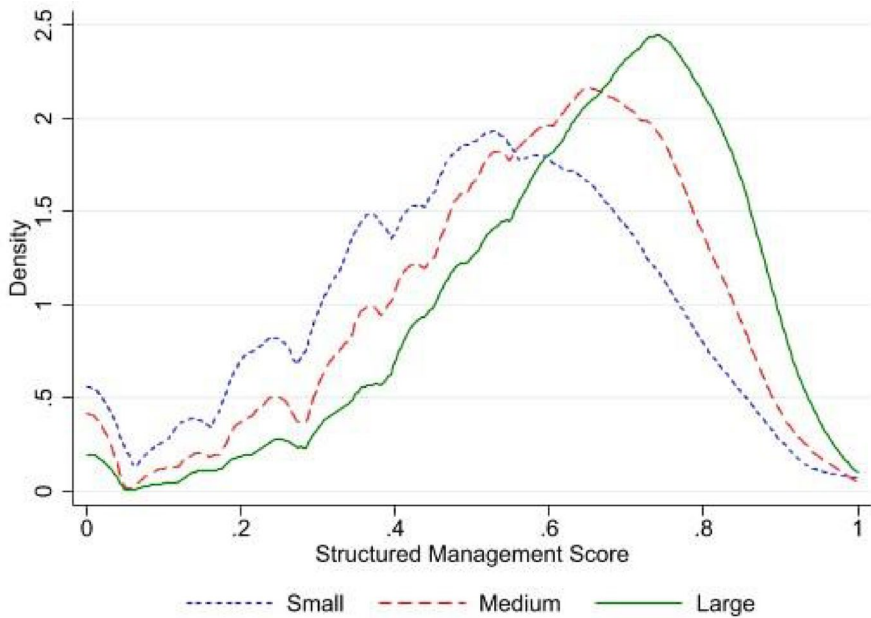


Fig. 2 Distribution of structured management scores by firm size

Table 2 Summary statistics

Indicator	<i>N</i>	Mean	Std Dev	Min	Median	Max
Sales	20,791	8885	22,027	1.83	1872	415,092
Capital	20,791	3852	12,270	0	523.87	141,183
Material	20,791	4359	9883	0	587.01	116,301
Firm age	20,791	26.85	21.11	1	22	223
Employees	20,791	136.39	853.12	1	50	115,810
Manager experience	20,791	21.38	11.92	1	20	70
Exporting	20,791	0.44	0.50	0	0	1
Training	20,791	0.68	0.47	0	1	1
Public	20,791	0.07	0.25	0	0	1
R&D	20,791	0.19	0.39	0	0	1
Foreign	20,791	0.11	0.32	0	0	1
SMP	20,791	0.57	0.19	0	0.59	1
Monitoring	11,719	0.63	0.23	0.33	0.67	1
Incentives	19,348	0.53	0.32	0	0.50	1
Targeting	20,433	0.49	0.31	0	0.58	1
Operations	19,200	0.73	0.27	0	0.67	1

Authors' analysis based on data from Enterprise Surveys (World Bank 2018–2023). Monetary values for sales, material, and capital in thousands of 2015 US dollars. Employees are full time employees

time employees respectively. Despite the large proportion of smaller firms, the data on sales, material and capital indicate that the firms handled large amounts of business. The median firm reported sales of approximately \$1,871,830 (in constant 2015 US dollars), but with a very high standard deviation. The average firm age was 27 years, indicating a predominantly older sample of firms. Of the surveyed firms, 44% reported Exporting during the fiscal year, and 11% had at least 10% foreign capital. Notably, over two thirds of firms allocated resources to employee training, and nearly 20% reported expenditures on R&D. Lastly, the average top manager had over twenty years of experience.

The estimation sample is substantially smaller than the full WBES because our production function specification requires non-missing values for sales, capital, materials, and all control variables. We examine whether firms in our estimation sample differ systematically from those excluded in Sect. 6.

5 Estimation methodology and outcomes

In this paper, we explore structured management practice's effect on firm performance as measured by sales revenues. The baseline regression involves estimating the following equation (Bloom and Van Reenen, 2007; Bloom et al., 2012b; Van Reenen, 2013; Broszeit et al., 2019; Grover and Karplus, 2020):

$$\ln(y_i) = \beta_0 + \beta_1 \ln(N_i) + \beta_2 \ln(K_i) + \beta_3 \ln(M_i) + \beta_4 Z_i + \beta_5 X_i + \eta_n + \nu_j + \gamma_t + \varepsilon_{icjt}$$

where subscripts i, n, j , and t represent firm, country, industry, and year, respectively. The outcome variable, $\ln(y_i)$ denotes the log of firm sales measured in constant US dollars. $\ln(N_i)$ represents the natural logarithm of the number of employees in firm i , what we call the firm's size. Similarly, $\ln(K_i)$ and $\ln(M_i)$ denote the natural logarithm of capital and material in firm i , respectively. Z_i denotes the firm's management practice indicator, which could be the overall z -management practice score or its four component categories as described earlier. A positive and statistically significant estimate of β_3 indicates that firms with higher management scores tend to exhibit higher sales revenues.

X_i is a vector of other firm-level control variables known to influence performance, including the log of firm age, log of manager experience, and dummy variables indicating whether a firm has an in-house training program, whether it is publicly traded, conducts R&D, exports, and has a significant degree of foreign ownership. The model also includes controls for country (η_n), industry (ν_j), and year (γ_t) fixed effects. Country fixed effects account for cross-country variations in institutions and governance that can impact firms' performance. Industry fixed effects help to adjust for differences in technology and organizational capabilities across different industries. Year fixed effects control for economic and social conditions that may influence sales for the sample in a given year. Although our estimations do not apply sampling weights, the extensive inclusion of fixed effects aligned with the survey stratification variables mitigates concerns that omitting sampling weights would bias the estimated conditional relationships.

Standard errors are clustered at the country level to account for within-country correlation of errors (MacKinnon et al., 2023). Standard regression models assume each observation is independent, but firms within the same country are subject to similar country-level shocks, creating correlated error terms. Failing to account for this correlation would yield artificially small standard errors and overstated statistical significance.

5.1 The overall effect of SMP: baseline results

In this section, we begin exploring the correlation between structured management practices (SMP) and firm revenues. For ease of interpretation, SMP and its components are measured on a standardized scale (i.e. $z \sim N(0, 1)$).

Baseline regression results are provided in Table 3. The table presents a set of OLS regressions starting with a parsimonious specification, then progressively adds control variables. All specifications use the same sample of 20,791 observations to facilitate comparison of coefficients across columns. For the first two specifications, the relationship between the use of management practices and firm performance is positive and statistically significant. Column (1) shows the correlation between the

Table 3 Baseline regression results

	Dependent variable: log (sales)		
	(1)	(2)	(3)
SMP	0.543*** (0.035) [0.000]***	0.071** (0.033) [0.074]*	0.014 (0.009) [0.200]
log (employees)		0.859*** (0.032)	0.487*** (0.035)
log (capital)		0.247*** (0.016)	0.080*** (0.01)
log (material)			0.464*** (0.037)
log (firm age)			0.031* (0.017)
log (MGR experience)			0.018** (0.009)
Exporting			0.100*** (0.021)
Training			− 0.051*** (0.019)
Public			0.109*** (0.023)
R&D			0.003 (0.020)
Foreign			0.174*** (0.034)
Constant	13.811*** (0.224)	6.996*** (0.279)	5.009*** (0.328)
Country FE's	Yes	Yes	Yes
Year FE's	Yes	Yes	Yes
Industry FE's	Yes	Yes	Yes
Observations	20,791	20,791	20,791
R-squared	0.352	0.761	0.872
Adjusted R-squared	0.349	0.76	0.872

Column (2) adds logarithms of capital and labor, approximating a standard neoclassical production function while Column (3) reports estimates from the full regression model, which includes all remaining control variables. SMP is measured as a z-score. Clustered robust standard errors at country level are in parentheses. Wild bootstrap p-values are in square brackets below standard errors; calculated using Webb (2014) six-point weights, 999 replications, clustered by country. See MacKinnon et al. (2023). Authors' analysis based on data from Enterprise Surveys (World Bank 2018–2023)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

management practices score and sales, including only fixed effects for country, year, and industry. It indicates a very strong positive association between better management practices and firm sales. For every one-standard deviation increase in SMP, sales are higher by 54.3%. Adding covariates decreases this value, probably due to SMP's correlation with the covariates.

Column (2) adds logarithms of capital and labor, mimicking a standard neoclassical production function. The management coefficient remains positive and significant. The estimated coefficient of 0.071 indicates that firms which score one standard deviation higher on the management practice score are expected to achieve approximately 7.1% higher sales.

The final column of Table 3 adds controls for the value of materials, and other firm level controls. The management coefficient remains positive albeit smaller (0.014) and is no longer statistically significant. This suggests that much of the raw correlation between SMP and sales operates through channels captured by our control variables: firm size, capital, and materials. In practical terms, firms with a one standard deviation higher management score achieve modestly higher sales revenues, though this direct effect is not precisely estimated in the full specification.

Regarding firm characteristics, firms with more employees, materials, and capital have much higher sales. Additionally, publicly traded firms and those engaged in international sales also demonstrate more sales. These findings align with research conducted at the country level (Bloom et al., 2012b; Broszeit et al. 2019). Variance inflation factors for all substantive regressors in the full specification (column 3) range from 1.18 to 2.84, indicating that multicollinearity is not a concern.

Overall, these results support Hypothesis 1: structured management practices are positively associated with firm performance. Wild bootstrap p-values are consistent with the conventional results in columns (1) and (3). In column (2), the bootstrap p-value of 0.074 suggests significance at the 10% level rather than 5%, though the positive association remains.

5.2 The effect of SMP by country-level income groups

Cross-country studies that examine the link between management practices and firm performance have primarily focused on data from developed economies.² In contrast, this study uses a comprehensive database that includes more than fifty countries across different levels of economic, institutional, and social development. Specifically, the dataset includes data from nearly 250 thousand firms in 167 countries. Of these, 28% are classified as high-income, 43% are middle-income, and 28% are low or lower-middle income countries. Approximately 33% of the firms in the dataset are in high-income countries, 45% are firms in middle-income countries, and 21% are firms in low or lower income (i.e. developing) countries. Due to this high heterogeneity, analyzing only richer countries, or even the total dataset in aggregate, could obscure noteworthy differences in the correlation between SMP and sales.

² The exception is the study by Bloom et al. (2012b) that includes India and ten transition countries in Europe and central Asia.

Table 4 examines the relationship between firm sales revenues and management practices across three country groups. One initial finding is that the management score is positively and significantly associated with higher sales in low-income, and high-income countries. In fact, the impact of SMP is identical in developing countries as it is among high-income countries. Focusing on columns (1) and (3), firms with a one standard deviation higher management score achieve approximately 1.5% higher sales in lower and high-income countries.

Further, Table 4 shows that the marginal benefit of labor is higher for high-income countries, and the marginal benefit of capital is higher in low-income countries. This likely reflects the relative scarcity of capital among low-income countries, and the relative scarcity of labor in higher income countries. The impact of being publicly traded is also equal between high income and low-income countries, with no effect for firms in middle-income countries.

Irrespective of the country's income level, firms with foreign capital exhibit higher sales. Finally, exporting firms achieve higher sales across all three income categories. These results support Hypothesis 2: the positive association between structured management practices and firm performance extends to developing countries. The

Table 4 Results by country-level income groups

	Dependent variable: ln (sales)		
	Low & lower-mid	Upper-middle	High income
	(1)	(2)	(3)
SMP	0.015* (0.008) [0.0290]**	0.031 (0.020) [0.17]	0.015* (0.008) [0.114]
log (employees)	0.405*** (0.032)	0.509*** (0.042)	0.636*** (0.064)
log (capital)	0.077*** (0.012)	0.081*** (0.016)	0.055*** (0.010)
log (material)	0.548*** (0.030)	0.408*** (0.045)	0.337*** (0.051)
log (firm age)	0.040 (0.029)	0.035 (0.023)	0.031** (0.011)
log (manager experience)	0.011 (0.016)	0.021 (0.018)	0.004 (0.010)
Exporting	0.140*** (0.025)	0.114* (0.056)	0.092*** (0.027)
Training	− 0.083*** (0.020)	− 0.106** (0.045)	− 0.034** (0.015)
Public	0.132*** (0.033)	0.066 (0.062)	0.104*** (0.023)
R&D	0.015 (0.054)	− 0.016 (0.038)	0.040*** (0.019)
Foreign	0.183*** (0.056)	0.160*** (0.053)	0.229*** (0.049)
Constant	3.611*** (0.172)	5.897*** (0.335)	7.344*** (0.519)
Country FE's	Yes	Yes	Yes
Year FE's	Yes	Yes	Yes
Industry FE's	Yes	Yes	Yes
Observations	9,962	3,856	6,973
R-squared	0.868	0.835	0.866
Adjusted R-squared	0.868	0.833	0.865

Column (1) reports results for firms in low- and lower-middle-income countries; Columns (2) and (3) report results for firms in upper-middle- and high-income countries, respectively. SMP is measured as a z-score. Clustered robust standard errors at country level are in parentheses. Wild bootstrap p-values are in square brackets below standard errors; calculated using Webb (2014) six-point weights, 999 replications, clustered by country. See MacKinnon et al. (2023). Authors' analysis based on data from Enterprise Surveys (World Bank 2018–2023)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

point estimates for low- and high-income countries are nearly identical, though the association for high-income countries is less robust to the wild bootstrap. Wild bootstrap p-values corroborate the positive association for low- and lower-middle-income countries. The bootstrap p-value of 0.029 is in fact stronger than the conventional result. For high-income countries, however, the bootstrap p-value of 0.114 falls short of conventional significance thresholds, suggesting that finding should be interpreted with caution.

5.3 The effect of SMP by firm size

How does adopting good management practices vary across firm size? This relationship is particularly relevant because it has been asserted (without evidence) that “formal management practices may not be necessary” for smaller firms (Bloom et al., 2016, p.12). A similar opinion is evident in McKenzie and Woodruff (2017): “Management in the small-firm context is less focused on human relations than is the case in larger firms” (p. 2967).

The WBES classifies small firms as those with 5 to 19 employees and medium-sized firms with 20 to 99 employees. Of the 71,272 firms in the WBES, our data source, small firms account for about 21% of the total.

A look back at Table 1 reveals that the proportion of firms using structured management practices increases with firm size. Small firms do indeed adopt structured management practices ($SMP=0.498$), but to a lesser extent than medium-sized (0.573) and large firms (0.644). Moreover, the adoption patterns across various dimensions of management practices are similar to those observed in medium and large firms. The pertinent question is whether management practices are associated with better firm performance for small (and medium) firms.

Table 5 reports regression results by firm size. To our knowledge, these results provide the first evidence of a positive and statistically significant relationship between structured management practices and sales for small firms. In fact, the association is strongest among small firms, followed by medium-sized firms. Small firms that with management scores one standard deviation higher are associated with 3% higher sales. For medium firms this number drops to 2%. For large firms, the effect is statistically insignificant. This pattern directly contradicts the assumption that structured management practices matter less for smaller firms.

Regarding firm characteristics, Table 5 reveals that firms of all sizes with foreign ownership and those engaged in international sales, exhibit higher revenues, underscoring the enhancing effect of foreign ownership on firm performance.

The fact that the marginal benefit of SMPs is highest among smaller firms, while uptake is smallest, suggests that smaller firms have not incorporated SMPs, not because of a lack of benefits, but because the cost of implementation may be prohibitive.

Overall, structured management practices are important for small firms. This suggests that governments can incentivize their adoption, or decrease the regulatory burdens that inhibit adoption, the latter of which is a theme stressed by Bloom et al. (2019). These results support Hypotheses 3 and 4. The positive association between structured management practices and firm performance extends to small firms, and

Table 5 Results by firm size

	Dependent variable: ln (sales)		
	Small firms	medium firms	Large firms
	(1)	(2)	(3)
SMP	0.030*** (0.008) [0.001]***	0.020* (0.011) [0.091]*	– 0.002 (0.021) [0.956]
log (employees)	0.515*** (0.052)	0.518*** (0.044)	0.432*** (0.037)
log (capital)	0.088*** (0.011)	0.082*** (0.013)	0.069*** (0.015)
log (material)	0.491*** (0.045)	0.464*** (0.040)	0.444*** (0.040)
log (firm age)	0.039* (0.021)	0.026 (0.017)	0.042 (0.031)
log (manager experience)	0.015 (0.011)	0.003 (0.014)	0.030* (0.018)
Exporting	0.145*** (0.033)	0.108*** (0.029)	0.085** (0.036)
Training	– 0.045* (0.025)	– 0.053* (0.028)	– 0.064** (0.029)
Public	0.082 (0.085)	0.083*** (0.028)	0.160*** (0.032)
R&D	0.005 (0.038)	–0.007 (0.020)	0.029 (0.029)
Foreign	0.114* (0.067)	0.173*** (0.046)	0.196*** (0.040)
Constant	5.099*** (0.426)	4.732*** (0.304)	5.825*** (0.384)
Country FE's	Yes	Yes	Yes
Year FE's	Yes	Yes	Yes
Industry FE's	Yes	Yes	Yes
Observations	4,234	9,403	7,154
R-squared	0.872	0.801	0.758
Adjusted R-squared	0.869	0.799	0.754

Column (1) presents regression estimates for small firms only, whereas Columns (2) and (3) report estimates for medium and large firms, respectively. SMP is measured as a z-score. Clustered robust standard errors at country level are in parentheses. Wild bootstrap p-values are in square brackets below standard errors; calculated using Webb (2014) six-point weights, 999 replications, clustered by country. See MacKinnon et al. (2023). Authors' analysis based on data from Enterprise Surveys (World Bank 2018–2023)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

this association is strongest among small firms. Wild bootstrap p-values fully corroborate the conventional results across all three size categories, reinforcing the finding that the positive association between structured management practices and sales is concentrated among smaller firms.

5.4 The effects of SMP's components

This study follows Bloom and Van Reenen (2007) in segmenting structured management practices into four categories: monitoring, incentives, targets, and operations. Country level scores differ significantly across these categories (Bloom et al. 2012a).

Table 6 examines the correlations between the four dimensions of SMP and firm sales for groups based on country income and firm size. The high degree of correlation between these four components makes isolating their unique contributions challenging. Most coefficients are statistically insignificant. Monitoring and Targeting are more important for firms in high-income countries. Monitoring and Incentives are more important among smaller firms than larger ones.

Table 6 SMP sub-scores and firm revenues by type of aggregation

	Dependent variable: ln (sales)					
	Country income groups			Firm size groups		
	Low & Lower-Mid	Upper-Mid	High	Small	Medium	Large
	(1)	(2)	(3)	(4)	(5)	(6)
Incentives	0.005 (0.022) [0.871]	0.040 (0.029) [0.234]	0.024* (0.012) [0.087]*	0.056** (0.025) [0.022]**	0.007 (0.012) [0.553]	0.001 (0.019) [0.950]
Targeting	0.013 (0.023) [0.625]	0.033 (0.025) [0.181]	- 0.013 (0.013) [0.315]	0.045*** (0.016) [0.011]**	0.007 (0.016) [0.662]	- 0.029 (0.020) [0.283]
Operations	- 0.013 (0.024) [0.604]	- 0.005 (0.023) [0.822]	0.031* (0.017) [0.115]	- 0.000 (0.027) [0.992]	0.007 (0.017) [0.718]	- 0.014 (0.029) [0.709]
	0.005 (0.008) [0.536]	- 0.003 (0.023) [0.908]	0.006 (0.014) [0.670]	0.027 (0.021) [0.308]	- 0.003 (0.012) [0.758]	0.028 (0.019) [0.140]
[Other control variables omitted for space]						
Country FE's	Yes	Yes	Yes	Yes	Yes	Yes
Year FE's	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE's	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,187	2,216	4,218	1,354	4,814	4,453
R-squared	0.874	0.833	0.871	0.852	0.836	0.784

Clustered robust standard errors at country level are in parentheses. Wild bootstrap p-values are in square brackets below standard errors; calculated using Webb (2014) six-point weights, 999 replications, clustered by country. See MacKinnon et al. (2023). Authors' analysis based on data from Enterprise Surveys (ES), World Bank 2018–2023

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Monitoring, Incentives, Targeting and Operations are reported as z-scores

Factors that are complementary to structured management practices also positively affect a firm's sales. Firms with foreign capital tend to achieve greater sales. Giorcelli (2021) has argued that the presence of foreign capital promotes effective management practices.

6 Limitations of the study

This study incorporates a comprehensive set of primary controls commonly used in related research. The results are consistent across different specifications, indicating robust correlations between management practices and firm characteristics in various settings. However, the estimated coefficients we've discussed above should be interpreted only as correlations, not causally.

Two types of endogeneity may affect our estimates. First, firms with better managers likely adopt more effective management practices and engage in other sales-enhancing activities (Bloom et al., 2016). This creates potential omitted variable bias. Second, reverse causality may occur, where firms use higher sales revenues to fund the implementation of more modern management practices (Bloom & Van Reenen, 2007). Further research is necessary to properly establish causality.

Our analysis builds on Bloom et al. (2012a, 2016, 2019), but differs in two ways. First, we use the World Bank Enterprise Survey dataset which covers firms of varying sizes across more than 90 countries at different stages of development. This allows us to examine cross-sectional associations between management practices and firm revenues, particularly for small firms that are underrepresented in prior work. Second, Bloom et al. (2019) use quasi-experimental variation to identify the causal drivers of management practices among US firms. Our study does not claim causal effects. Instead, we prioritize external validity across diverse economic and institutional settings.

A further limitation concerns sample selection. Our full specification requires non-missing values for sales, capital, materials, and all control variables, reducing our sample from over 250,000 firms in the WBES to 20,791 in the estimation sample. Firms with complete data may differ systematically from those with missing values. For instance, larger or better-managed firms may maintain more complete records. To assess whether our results are driven by selection into this subsample, we re-estimate the model excluding materials, which allows a larger sample of 21,854 firms. Results are qualitatively similar (see Appendix Table A4). Similarly, re-estimating the model excluding capital (which has a low response rate in the WBES) yields a sample of 24,550 firms with qualitatively similar results (see Appendix Table A5).

Another way to establish causality is to exploit a panel dataset and examine how firms' performance changes when they adopt SMPs. Access to a longitudinal dataset would be highly valuable even if limited to a small sample of countries. Unfortunately, the WBES is not a panel dataset. The WBES dataset might also be supplemented with questions about the duration for which firms have implemented these practices, their organizational designs, and the perceived constraints and benefits associated with their adoption.

7 Discussion and conclusion

This paper has examined the association between structured management practices and firm performance using the World Bank's Enterprise Surveys, a large dataset covering firms of varying sizes across 94 countries. It not only provides firsthand evidence on the adoption of these practices but also offers valuable insights into the relationship between adoption and its relationship to firm performance.

To summarize our findings with respect to our four hypotheses: we find support for all four. Structured management practices are positively associated with sales (Hypothesis 1). This association extends to firms in developing countries (Hypothesis 2) and to small firms (Hypothesis 3). The association is strongest for small firms (Hypothesis 4), contradicting the assumption that structured management practices are less relevant for smaller enterprises.

We find that there is significant heterogeneity in adopting these practices across and within countries. Using OLS, we present evidence that firms which adopt structured management practices achieve higher revenues. This holds true across a broad sample of countries with diverse cultural, social, and economic backgrounds and among firms of various sizes.

The findings reaffirm the benefits of adopting good management practices on firm performance. For instance, our study confirms the impact of management practices on sales revenues, as highlighted by Bloom and Van Reenen (2007), Broszeit et al. (2019) and Bloom et al. (2022). Additionally, we support the findings of Bloom and Van Reenen (2007) and Broszeit et al. (2019) regarding the heterogeneous effects of the four categories of management practices on firm performance. Finally, our findings emphasize the importance of management experience and the presence of foreign capital as complementary factors that enhance firm performance.

The main empirical contribution of this paper is to determine whether the correlation between modern "structured management practices" and higher firm performance extends to smaller firms, and firms in less developed countries. Our study finds that structured management practices have a strong and positive association with performance in firms of all sizes and in countries at all stages of development.

More importantly, the association between structured management practices and firm performance appears strongest among smaller firms, and firms in developing countries, two demographics that have been neglected in the literature thus far.

Most importantly, we shed some light on a persistent question in the "economics of management" literature: if modern management techniques are so valuable, why don't more firms use them? Is it a problem with small benefits, or large costs? We find that management has the largest positive impact on small firms, compared to medium and large firms. This, combined with the fact that smaller firms tend to use these techniques the least suggests that low marginal benefits are unlikely to explain non-adoption. Instead, high implementation costs, possibly including regulatory costs as emphasized by Bloom et al. (2019), may be the more relevant barrier. At the same time, we must acknowledge a key limitation: while the marginal benefit of adopting management practices appears largest for small firms, we lack data on the actual costs of adoption for these firms. Without cost information and causal identification

between adoption and firm performance, we cannot definitively distinguish among potential explanations.

These results have policy implications. Performance differences across firms and countries persist even after accounting for differences in technology, equipment, and capital. Management practices appear to be an important part of the explanation, and our findings support this across countries at all income levels and firms of all sizes. Policymakers in developing economies should consider incentives to promote adoption of structured management practices or help firms — especially small firms — identify and overcome implementation barriers.

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Declarations

Conflict of interest The authors have no conflicts of interest to declare.

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